SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – SIH26097**
- **Problem Statement Title- AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID- 140037**
- **Team Name (Registered on portal)- Hackspire 2..0**

Hackspire2..0

IDEA TITLE

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PROPOSED SOLUTION

SAHAYAK is a multilingual, voice-first AI assistant that understands the livelihood profiles of SC beneficiaries and maps them to NSQF-aligned skill development pathways and relevant, ongoing government schemes such as PM-AJAY, through Website and WhatsApp.

FEATURES

→ **Voice-First, Empathetic Interviews**

Regional-language conversation, no text forms. same flow now runs on both Website and WhatsApp.

→ **Live Government Scheme Layer**

Scheduled scraper & cache ensure that data on MyScheme.gov.in and PM-AJAY remain up-to-date; graceful fallback in case any source fails.

→ **Real Matching Engine**

Scoring using keyword weights, developed and validated locally – ranks NSQF occupations and skillsets with a confidence level.

→ **Dual-Channel: Website & WhatsApp**

One common backend supporting both website and WhatsApp users.

→ **GIA Admin Dashboard**

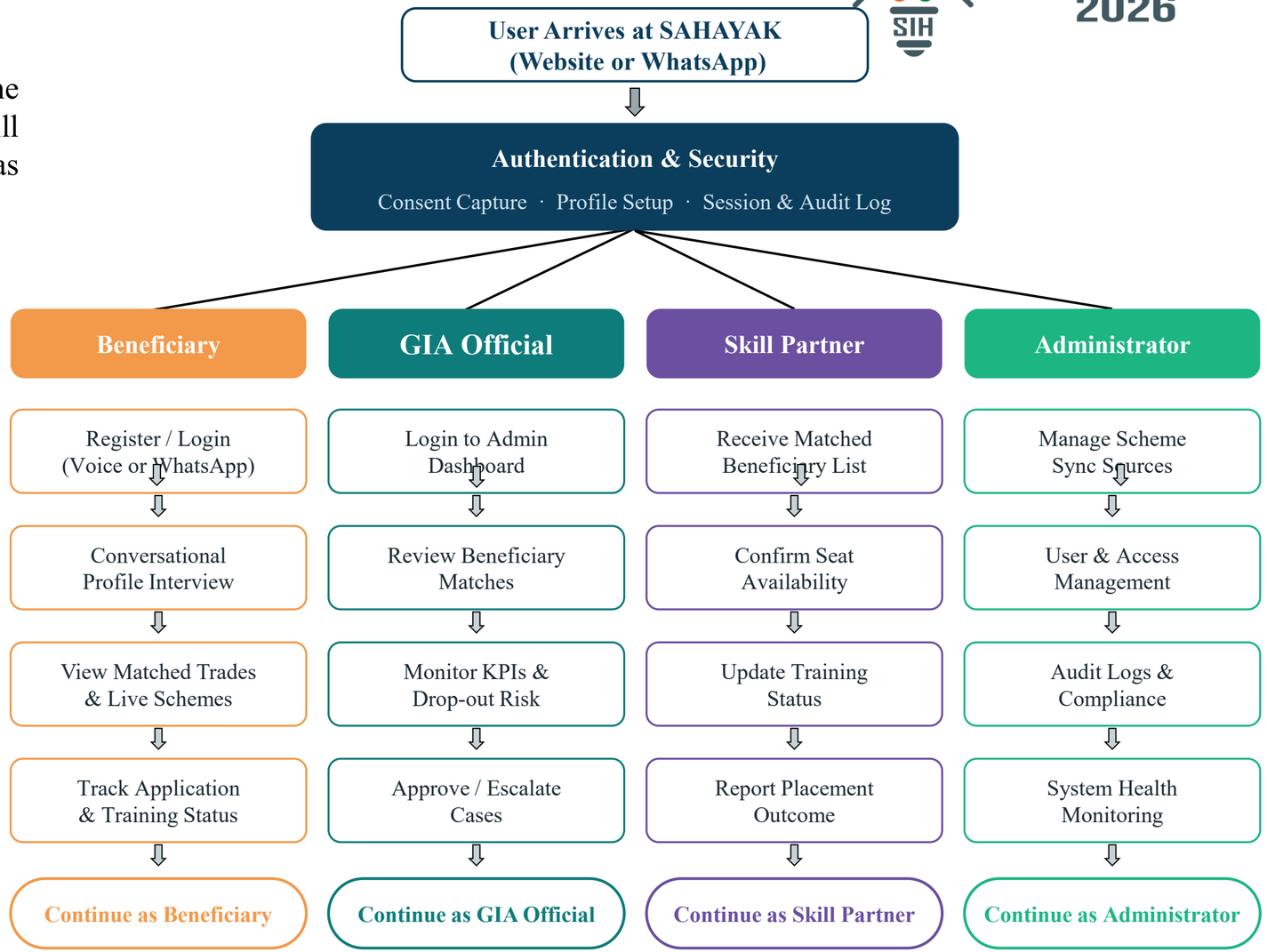
Live stats, channel split and beneficiary tracking pulled straight from the database.

→ **Grievance Support**

Enables beneficiaries to raise and track complaints related to caste-based discrimination through a simple, accessible interface.

→ **Explainable Semantic Matching Engine**

Maps dialect/synonym terms to NSQF trades & schemes via concept-expansion, explaining every match in plain language.



Sample match : input *"darzi"* → matched to **Tailoring & Fashion Design** (92%)
Why: 'darzi' maps to the Tailoring concept cluster; 3 active schemes found in your district.

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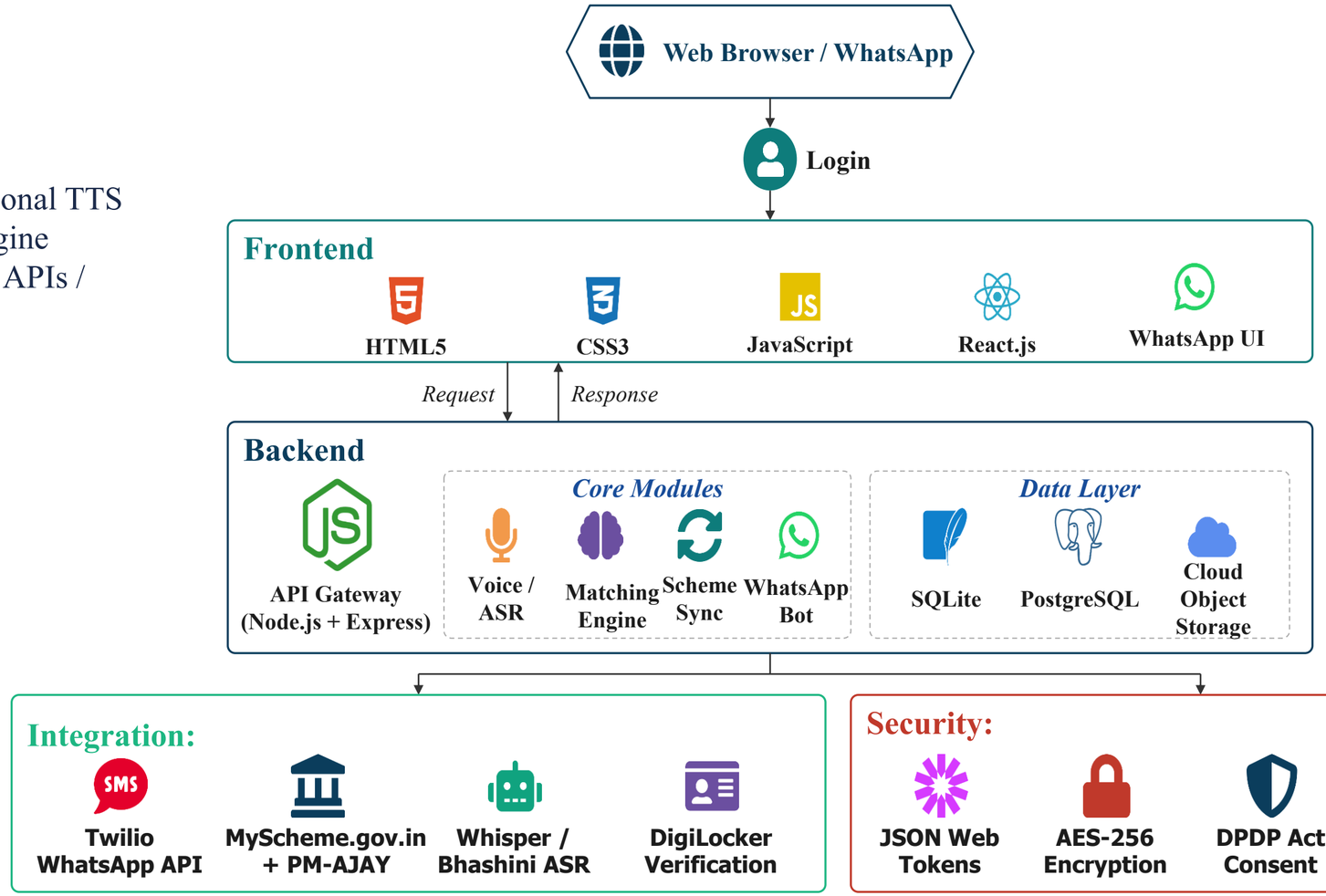
TECHNICAL APPROACH

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TECH STACK

- **Frontend:** HTML, CSS, JavaScript, React.js
- **Backend**: Node.js and Express.js
- **WhatsApp:** Twilio, WhatsApp API , N8N
- **Voice & Languages:** Bhashini , Whisper (ASR), Regional TTS
- **AI Matching:** Custom NSQF & Scheme Matching Engine
- **Live Scheme Data:** MyScheme.com, PM-AJAY Sync APIs / Scraper
- **Database:** PostgreSQL & Redis
- **Security:** JWT, AES-256, DPDP-compliant Consent Management
- **Cloud & Deploy**: AWS / MeghRaj, Docker
- **Admin Dashboard:** React.js & Chart.js

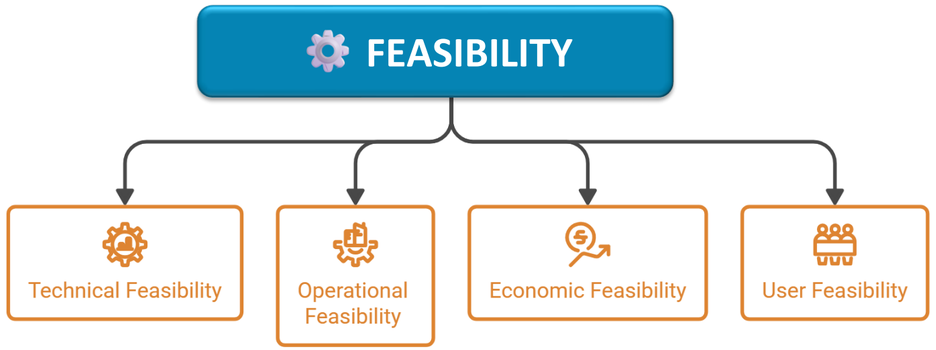




Live Prototype Link : click here

Youtube Link : click here

FEASIBILITY AND VIABILITY

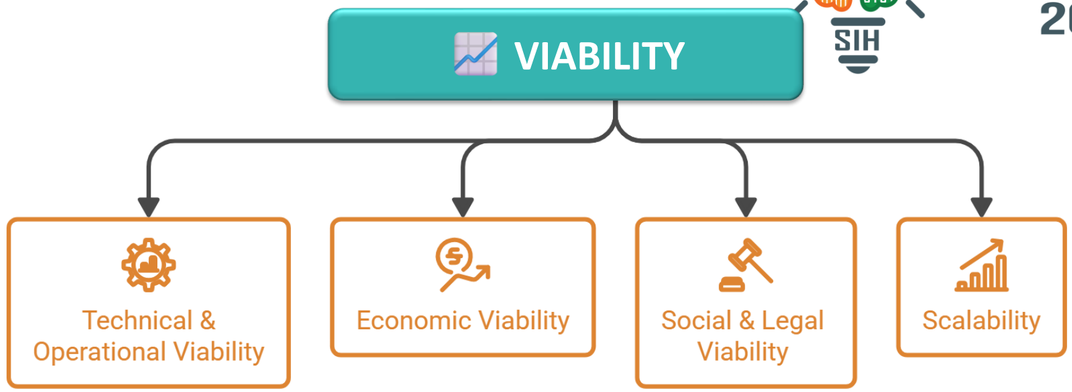


→ **Technical Feasibility**
- Uses **common and reliable technologies** like Voice AI, Backend, Database and WhatsApp.
- The system is divided into **separate modules**, making it easier to build and manage.

→ **Operational Feasibility**
- **Voice-based system** is easy to use and needs little or no training.
- Can be plugged into the existing **PM-AJAY GIA workflow** with minimal modifications

→ **Economical Feasibility**
- Utilizes **open source technologies** where possible and scales easily on **cloud infrastructure**.
- Saves on **cost** of human resources for **counselling**, **travel** and **documentation**.

→ **User Feasibility**
- **Voice-first design** is easy for rural and low-literacy users.
- Can be accessed via multiple channels: **Website**, **WhatsApp** reducing the barrier to access even for people who **do not have smart phones.**
- No typing or digital literacy required ; IVR planned next for users without smartphones



→ **Technical & Operational Viability**
- **Uses reliable, proven technology** to reduce risks and make long-term maintenance easier.
- **Automatic scheme data updates** reduce manual work like data entry.

→ **Economic Viability**
- **Automation reduces manpower needs**, keeping long-term costs low.
- **Fits within the existing PM-AJAY GIA budget**, reducing the need for new funding.

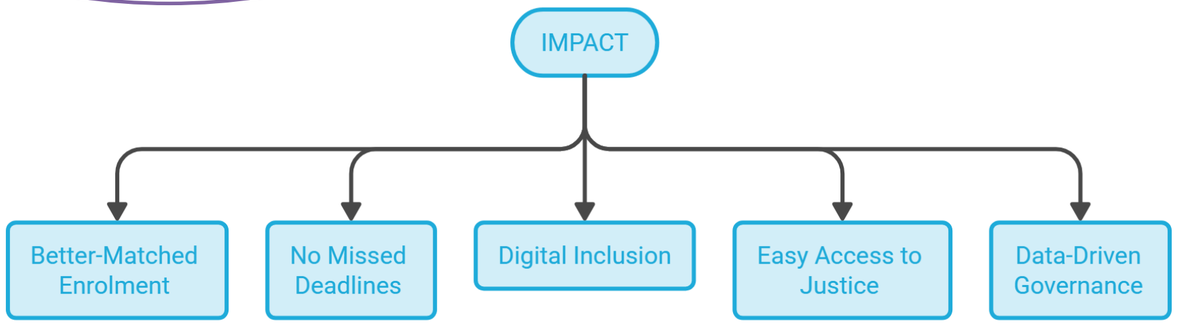
→ **Social & Legal Viability**
- Complies with **DPDP Act 2023** and **data protection** obligations such as consent, encryption, and data minimization.
- Features such as complaint filing operate within the **existing legal framework** of the **SC/ST (Prevention of Atrocities) Act 1989**, so no new regulatory approval is required.

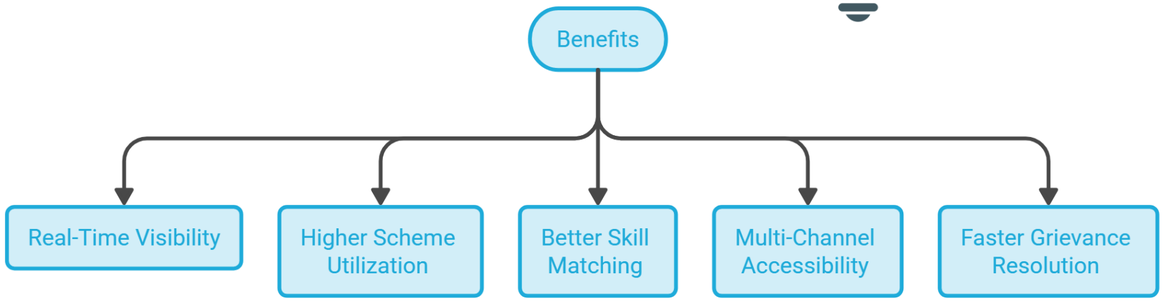
→ **Scalability**
- Can easily expand to **new districts and states**.
- No need to **redesign the complete system** for expansion.
- New channels like **kiosks and messaging apps** can be added easily.

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IMPACT AND BENEFITS

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IMPACT

BETTER – MATCHED ENROLMENT

People get training that matches their interests and aspirations.

NO MISSED DEADLINES

Live scheme updates help people stay informed about important deadlines.

DIGITAL INCLUSION

Voice, WhatsApp and (planned) IVR help people with low literacy or limited connectivity access services.

EASY ACCESS TO JUSTICE

One-click complaint filing makes it easier to report caste-based discrimination.

DATA – DRIVEN GOVERNANCE

The Ministry gets accurate, real-time data for better decisions.

BENIFITS

REAL TIME VISIBILITY

A single live dashboard provides instant enrolment, matching, and dropout data for faster decisions.

HIGHER SCHEME UTILIZATION

Auto-updated scheme information helps eligible users find and access active schemes.

BETTER SKILL MATCHING

Interest and demand-based matching reduces mismatched enrolments and dropouts.

FASTER GRIEVANCE SOLUTION

Integrated complaint routing makes reporting easier and speeds up formal action.

MULTICHANNEL ACCESSIBILITY

One platform through Website, WhatsApp, and IVR reaches users across different devices and literacy levels.

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RESEARCH AND REFERENCES

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TECHNOLOGIES RESEARCH

IndicWhisper (AI4Bharat / Bhashini)
Fine-tuned on Whisper models using 10,700+ hours of audio across 12 Indian languages.
Source: AI4Bharat / Vistaar (Interspeech 2023).
click here

HEEP-Indic
Trained on high-quality samples across 46 Indian languages for wider language accessibility.
Source: HEEP-Indic (2025 research).
click here

PRIMARY RESEARCH

7% of trained PMKVY candidates get placed, highlighting the skilling-to-employment gap.
- *Business Today*

35% of enrolled candidates dropped out in FY 2024–25, highlighting a growing challenge in skill training completion.
- *Business Today*

DATA GATHERED FROM PUBLIC

Scheme Awareness Survey from Engineering College of Nagpur
Click Here

DIGITAL & AWARENESS GAP

› DigiLocker adoption is below 25% in rural areas vs. 60% in urban areas, highlighting a major awareness gap.

› Rural internet penetration is only 37%, compared to 67% in urban India (TRAI, Dec 2022)

Features	SAHAYAK	Manual Counselling	Generic Chatbot Apps
Voice-first, regional-dialect interviews	☑	☑	✗
Live-refreshing govt. scheme database	☑	✗	✗
Real similarity-based eligibility matching	☑	✗	Keyword Only
WhatsApp + IVR feature-phone reach (Planned)	☑	✗	✗
Real-time GIA official dashboard	☑	✗	✗
Scales across districts without added manpower	☑	✗	✗